

Practice set 1

Powers, factorising and algebraic fractions

Write in the spaces

QUESTION ONE

(a) Simplify each expression, leaving your answer with positive indices.

(i) $\frac{(3y)^4}{3y^{-1}}$

(ii) $\sqrt[3]{8y^{27}}$

QUESTION THREE

(a) (i) Simplify fully $\sqrt{49y^{36}}$.

QUESTION ONE

(a) Solve $\frac{2x-3}{x+4} - 3 = 0$.

(b) (i) Factorise completely $6x^3y - 15x^2\sqrt{y}$.

(ii) Simplify fully $\frac{6x^2 - x - 12}{3x^2 - 5x - 12}$.

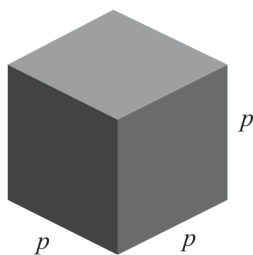
QUESTION TWO

(a) Simplify: $\frac{x^2 - x - 12}{4x + 12}$

(b) Write $\frac{5x}{x-3} - \frac{x-4}{x+2}$ as a single fraction in its simplest form.

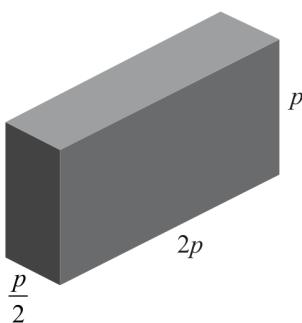
QUESTION THREE

Consider a cube with sides of p cm (where $p \neq 0$). The volume of the cube would be $p^3 \text{ cm}^3$, and the surface area of the cube would be $6p^2 \text{ cm}^2$.



Junyang wonders if it is possible to change the dimensions of the cube to make a cuboid that still has the same volume.

- (a) First, he tries doubling the length, halving the width, and keeping the height as p , as sketched below.

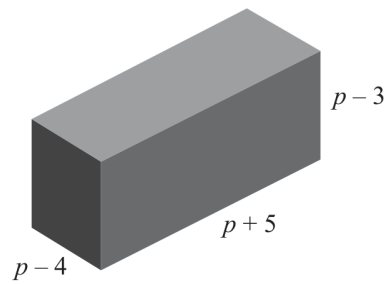


*Diagram is
NOT to scale*

Using algebra, find the volume of this cuboid.

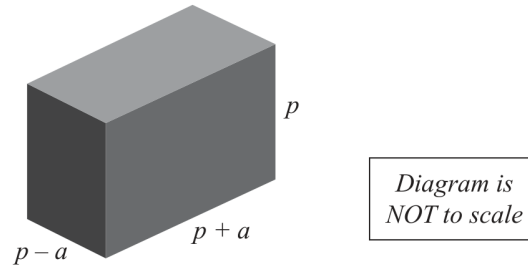
(b) The volume of the cuboid below is given by the expression: $(p - 4)(p + 5)(p - 3)$.

Expand and simplify this expression.



*Diagram is
NOT to scale*

- (c) Next, Junyang tries adding an amount, a , to the length and then taking off the same amount from the width, keeping the height the same (see below).



Are there any values of a for which the volume of the cuboid will be the same as the volume of the cube?